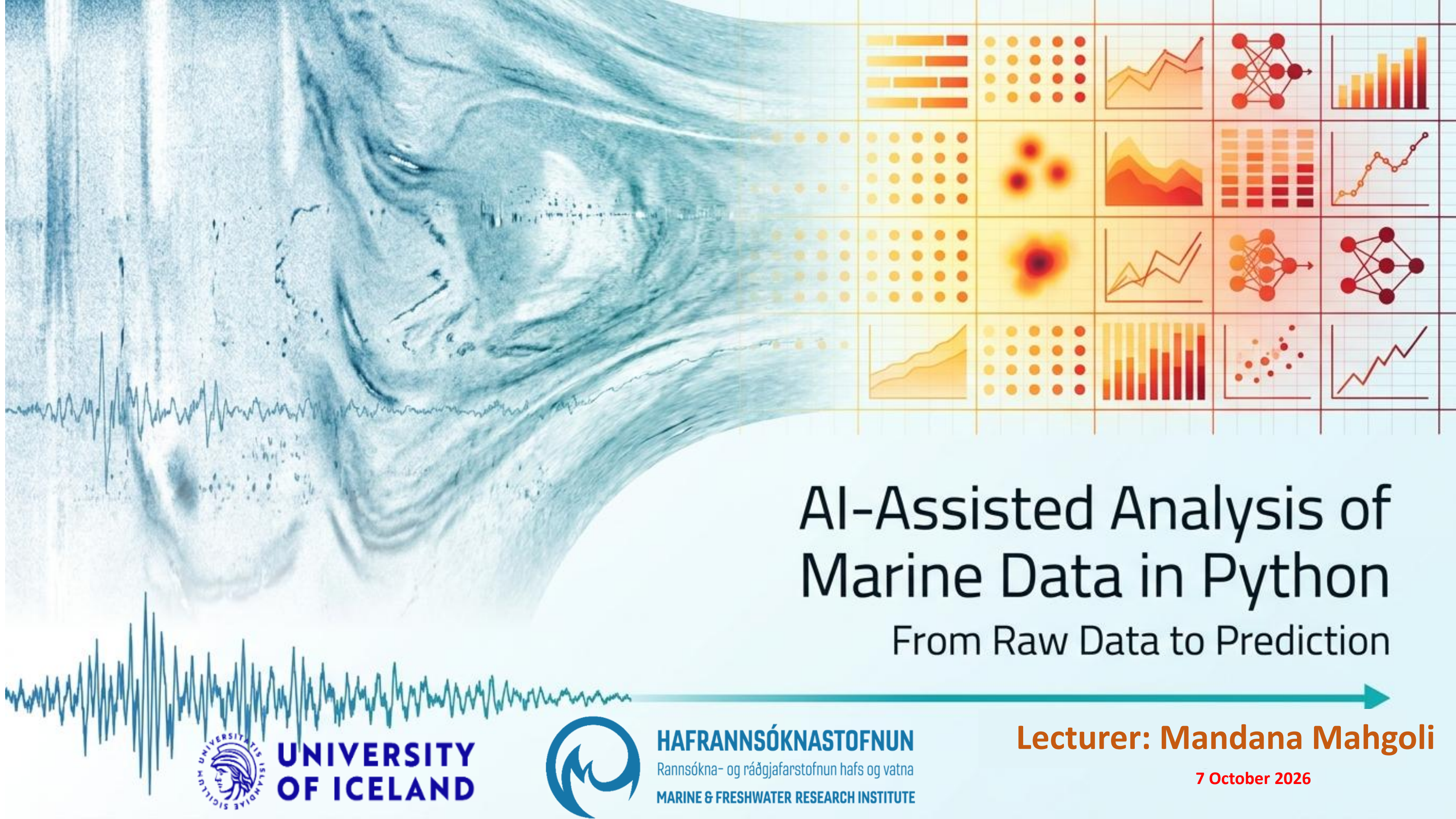




AI-Assisted Analysis of Marine Data in Python

From Raw Data to Prediction



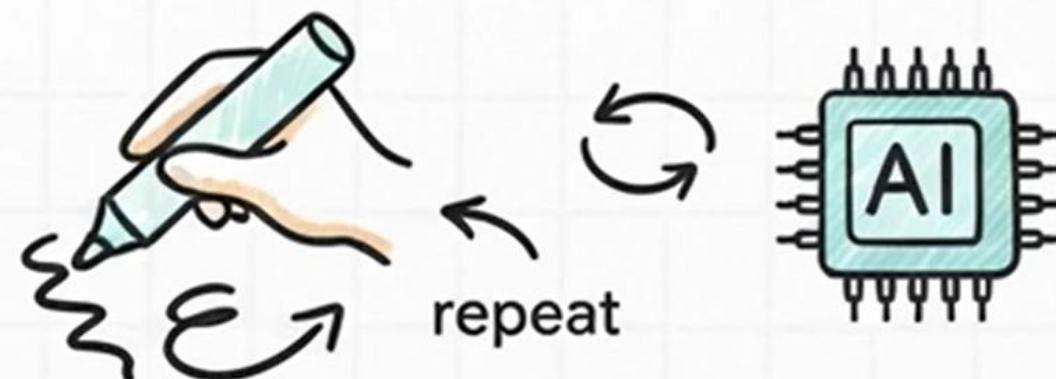
**UNIVERSITY
OF ICELAND**



HAFRANNSÓKNASTOFNUN
Rannsókn- og ráðgjafarstofnun hafs og vatna
MARINE & FRESHWATER RESEARCH INSTITUTE

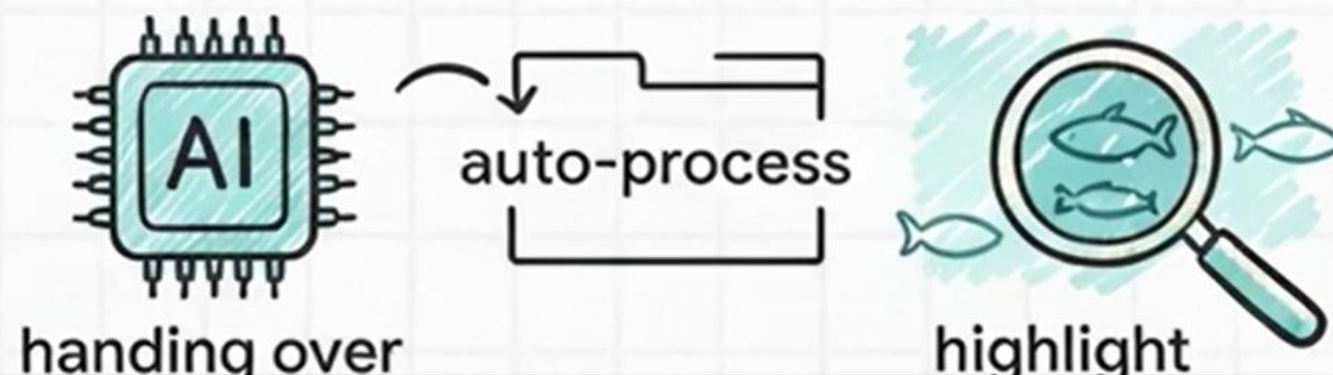
Lecturer: Mandana Mahgoli

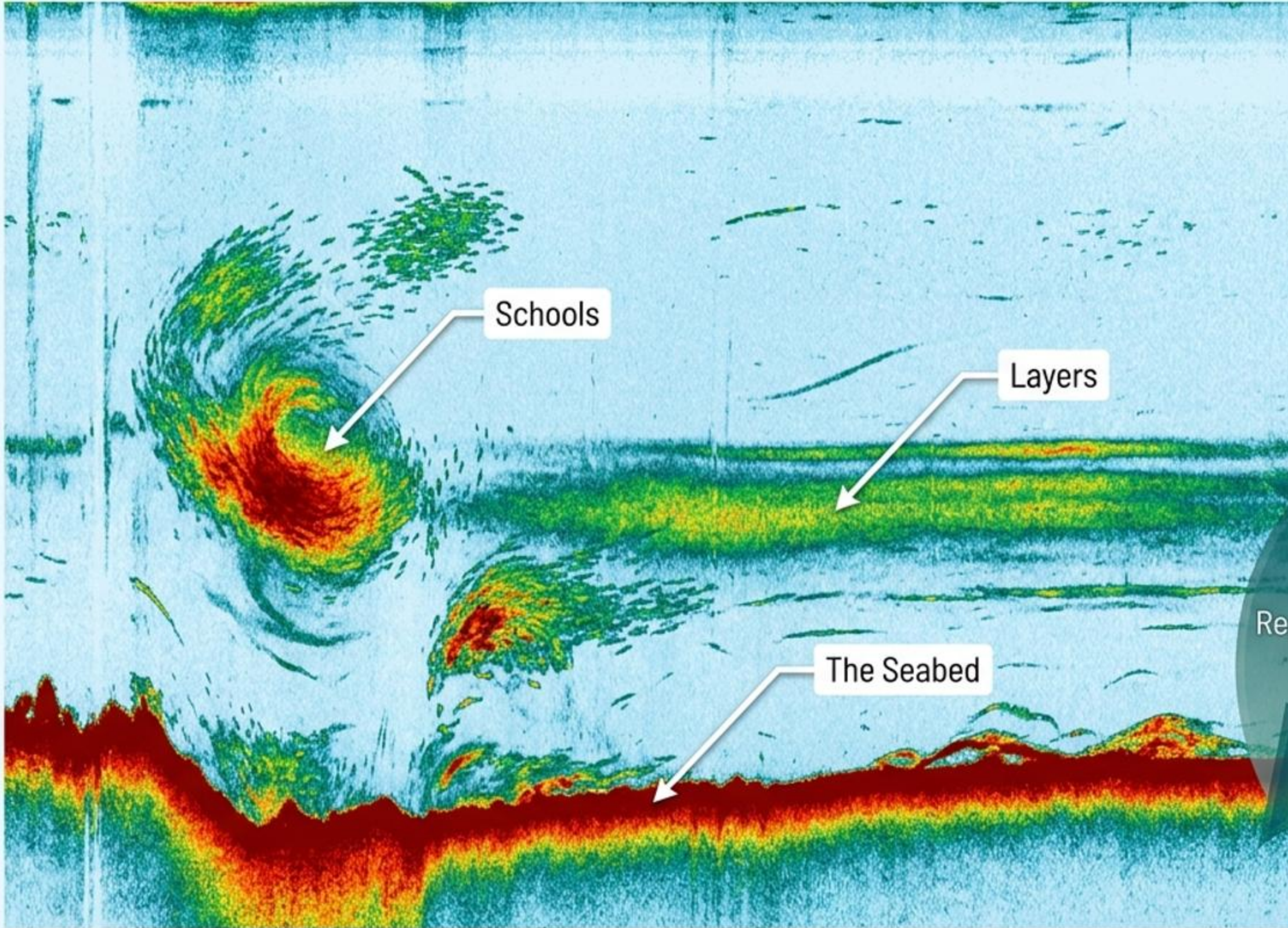
7 October 2026



How much **repetitive**
work could we
automate?

save

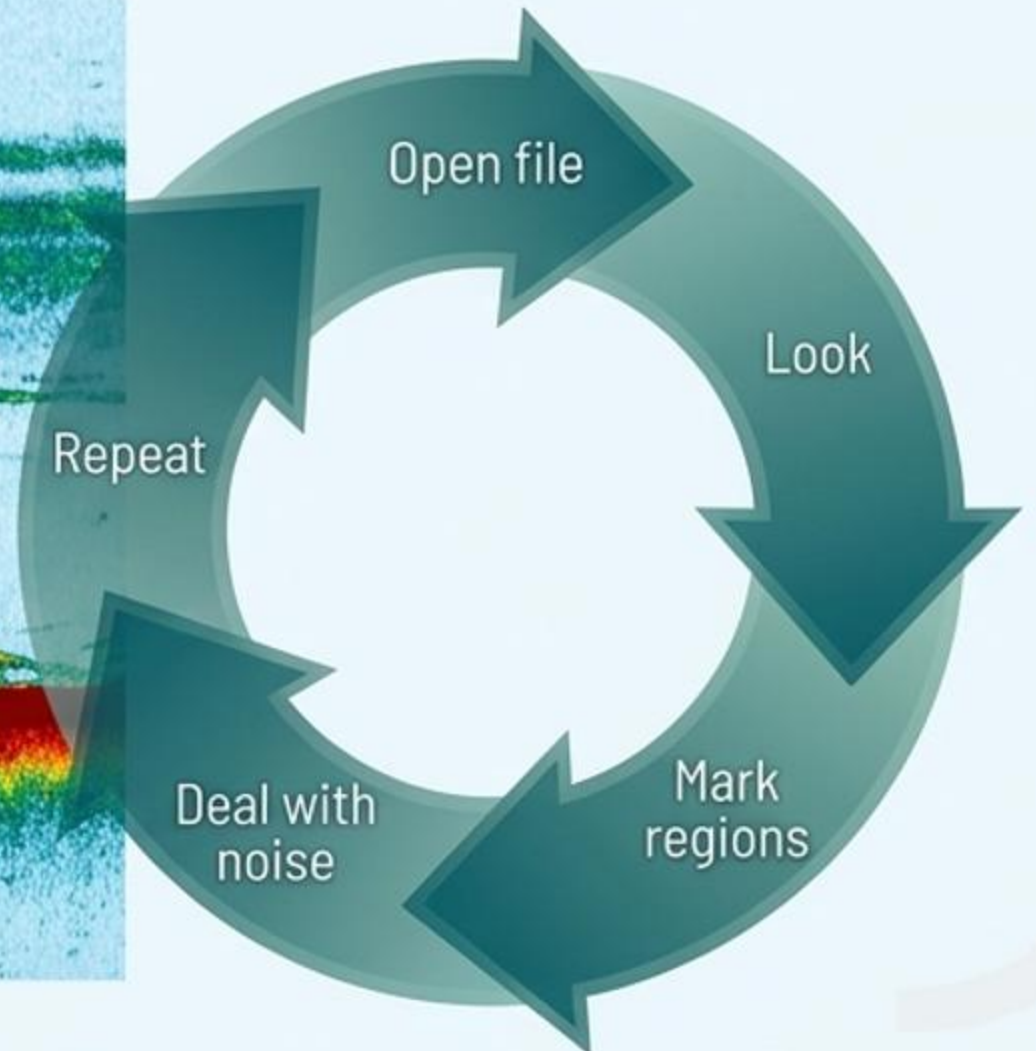




Starting With Data You Already Know

An echogram is a rich dataset. But the work around it is highly repetitive.

How much of this repetitive manual labor could we hand to a computer? Could we automatically find regions worth looking at, or simply make the data easier to work with?

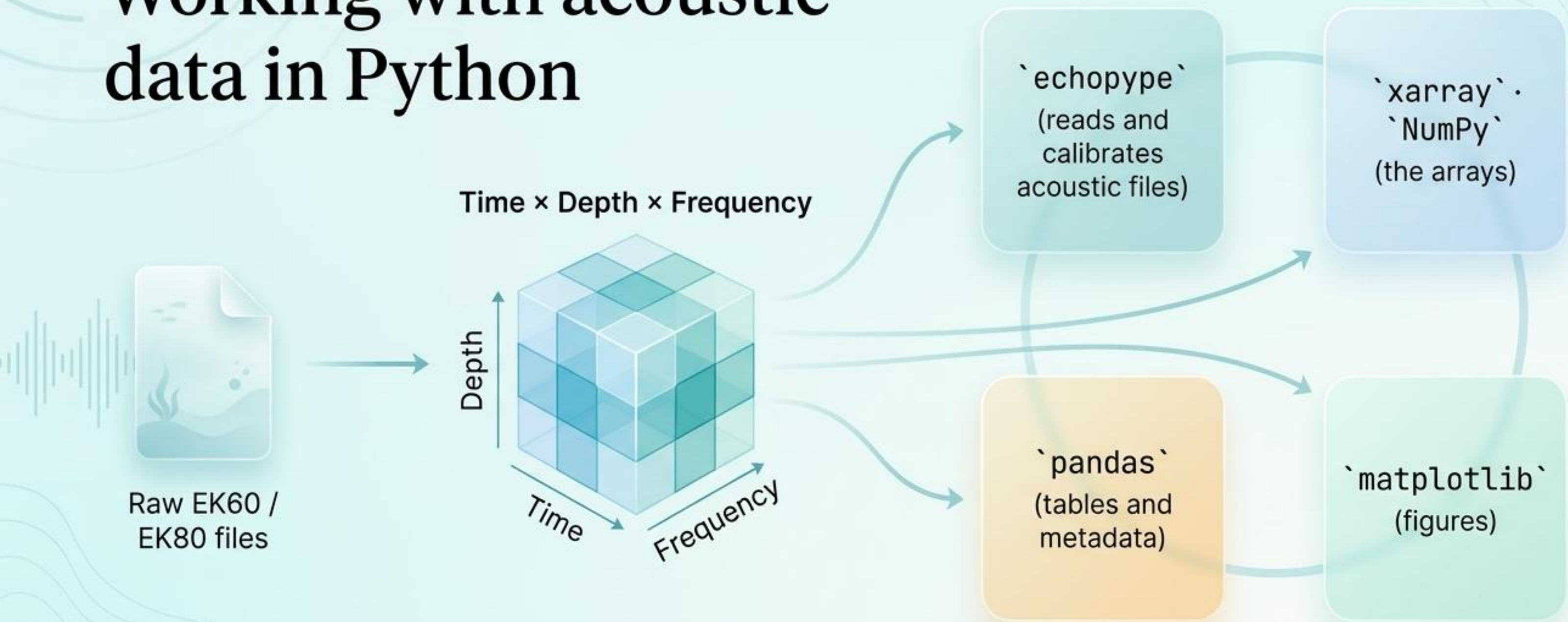


AI is not necessarily the first step.



Quite often the biggest gain comes from getting the data into a usable structure, plotting it properly, or automating one thing that's currently done by hand fifty times.

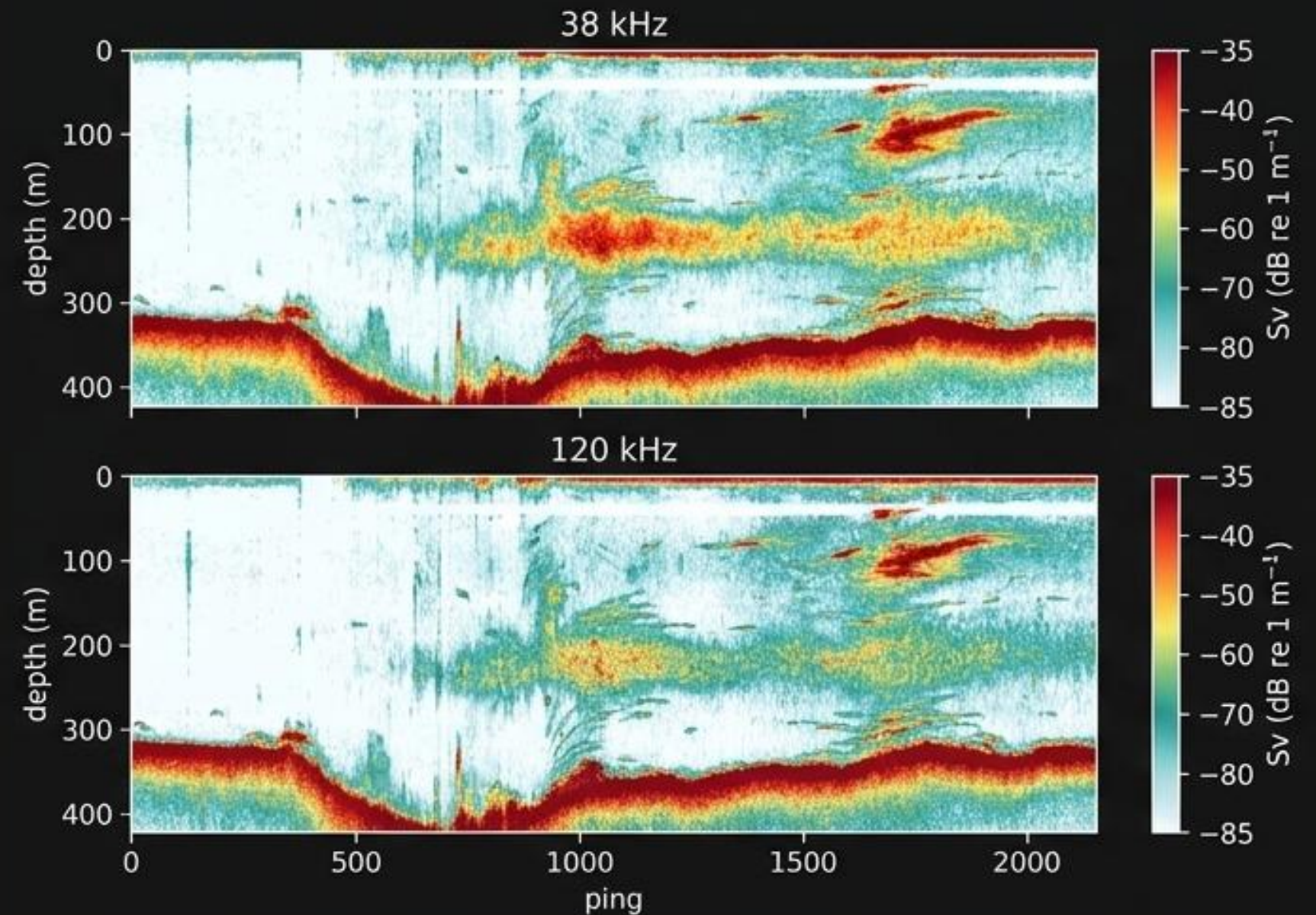
Working with acoustic data in Python



Making Data Visible via Natural Language

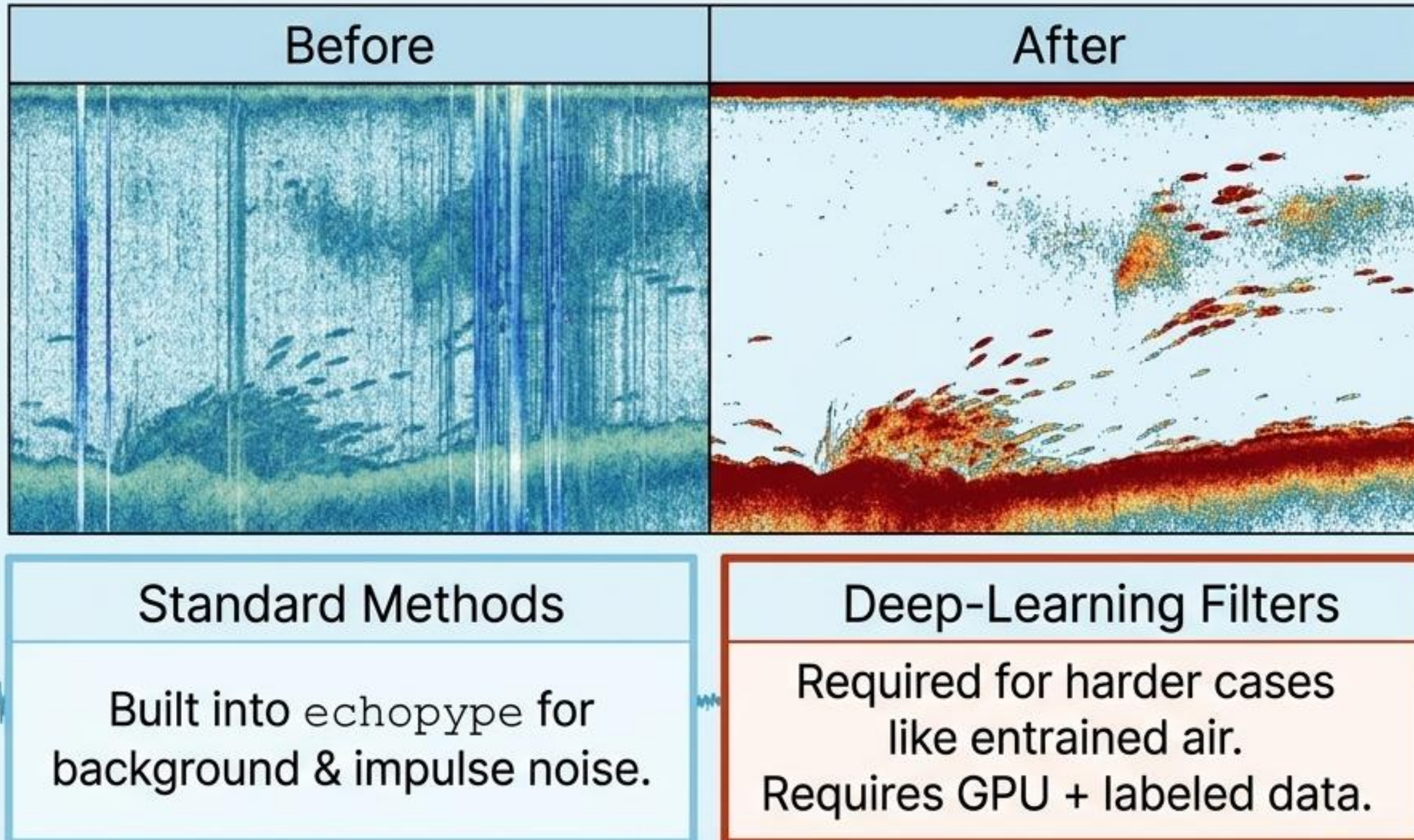
A few lines of code—written by the AI assistant, checked by the expert. A figure stops being decoration and becomes the first step of analysis.

Prompt: Plot the same transect at 38 kHz and 120 kHz using the standard ekmap colors. Limit vmin to -85 and vmax to -35.



Cleaning the Data: Standard vs. Deep Learning

AI doesn't clean the data for you. It helps find what deserves attention and writes the tedious code. What is scientifically meaningful remains your call.



Live example 1 — what I asked, what it wrote

What I asked the assistant *“Simulate a two-frequency echogram (38 and 120 kHz) with a noise floor that rises with range, a scattering layer, a few fish schools, a seabed and some impulse noise. Plot both side by side, then remove impulse and background noise.”*

```
Sv38, Sv120 = simulate_channel(38), simulate_channel(120)

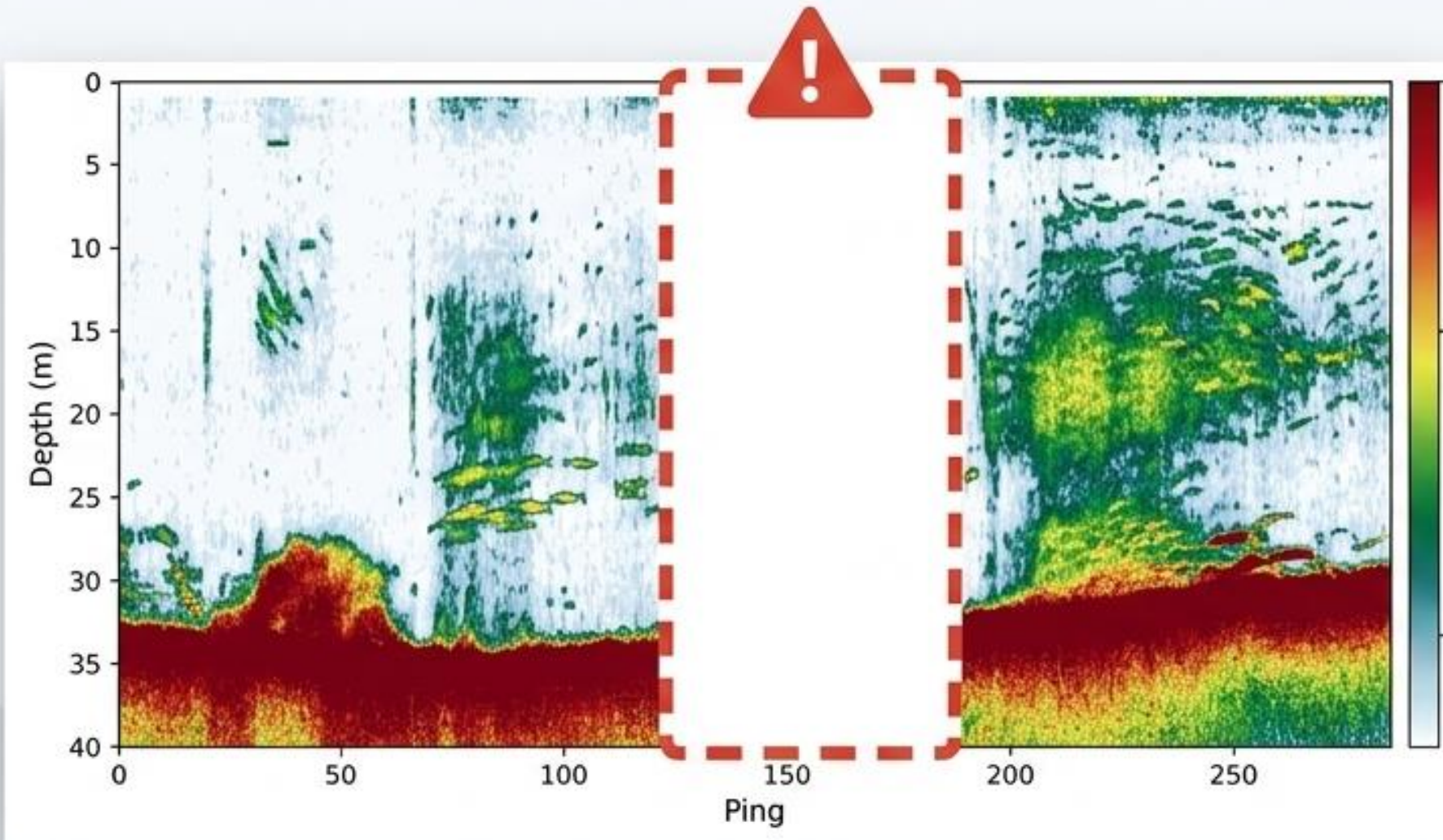
# the two standard cleaning steps (same methods as echotype)
Sv120_clean = remove_background_noise(
    remove_impulse_noise(Sv120),          # ping vs. neighbours' median
    depth, alpha=0.04,                    # absorption at 120 kHz, dB/m
    ping_num=20, range_sample_num=20,    # noise from the quietest block
    snr_threshold_db=10)                  # mask within 10 dB of noise

show_echogram(ax, Sv120_clean, depth, "120 kHz — after cleaning")
```

What I checked before trusting it

- absorption values — the assistant's first guess was wrong
- depth axis orientation (surface at the top)
- how many samples the SNR mask removed
- the schools and the seabed survived the cleaning

Code can run perfectly and still be wrong.



An assistant will happily give you a script that runs without a single error, produces a beautiful plot, and has quietly used the wrong column, the wrong units, or dropped half the pings. It won't warn you.



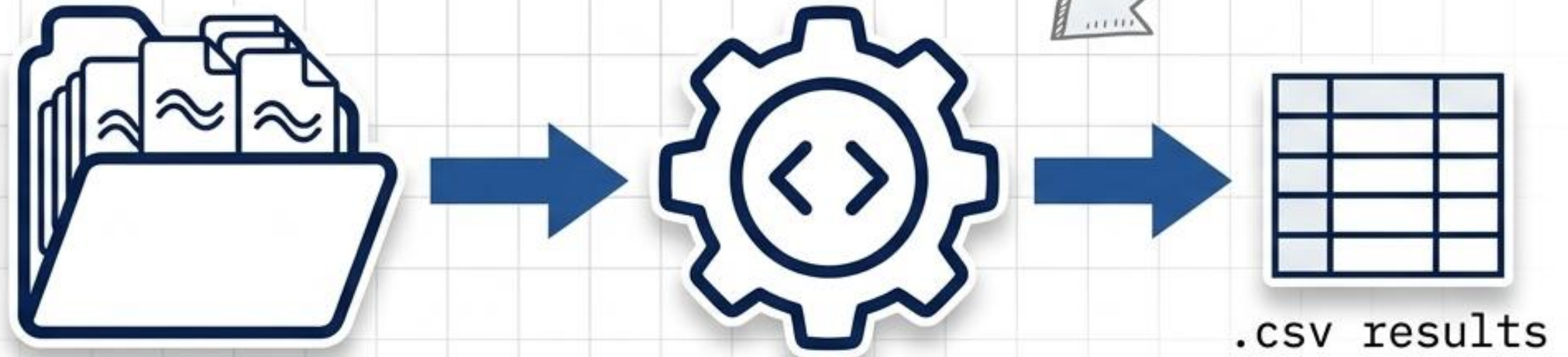
'The code ran' is never evidence that the analysis is correct. You look at the output and sanity-check it.

From one analysis to many.

Manual Workflow



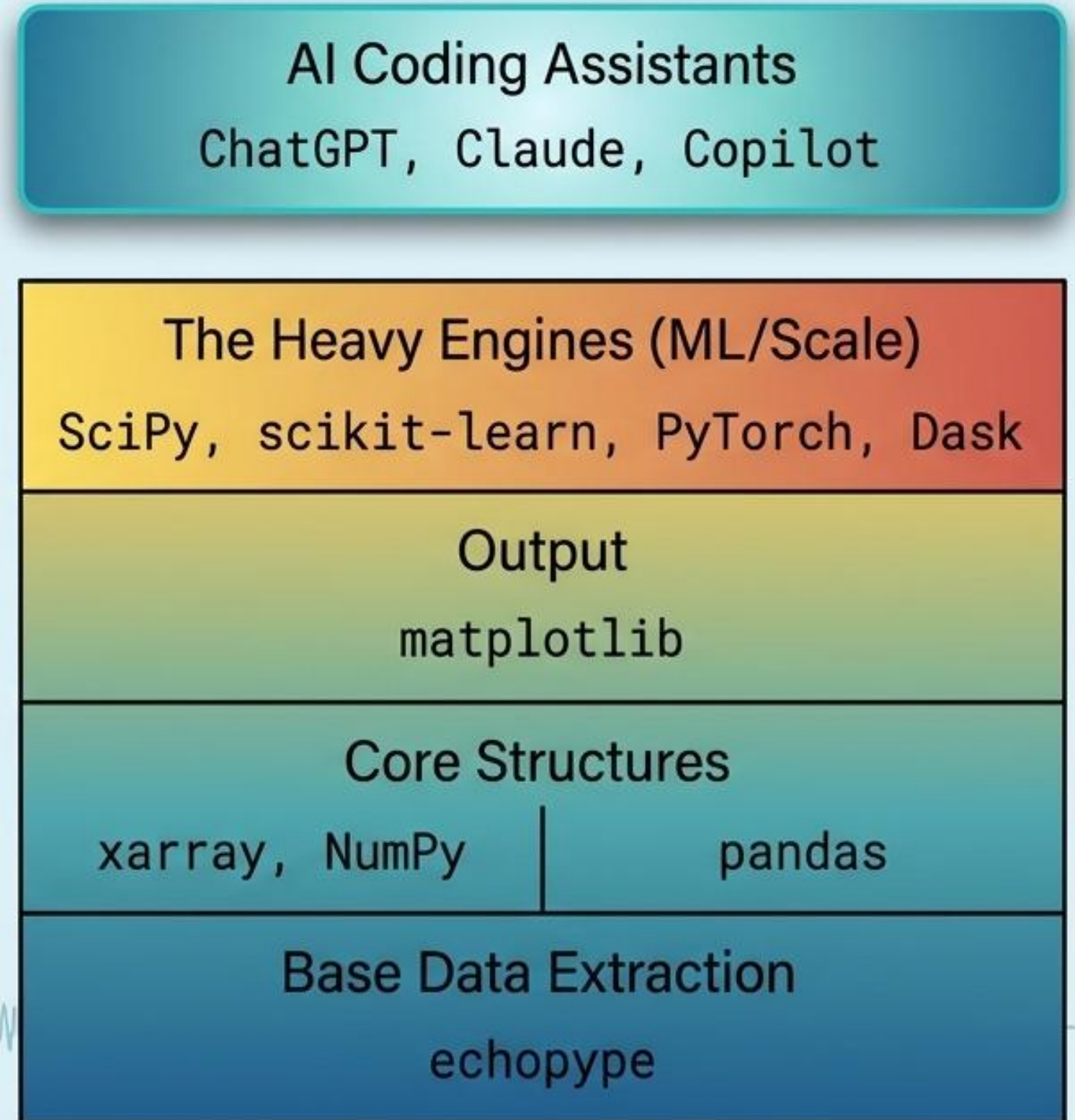
Python Pipeline Workflow



This is not really AI. It is just good computational practice—and often where the biggest time saving is found. The AI assistant excels at turning a script for one file into a loop for a folder.

The Marine Python Ecosystem

Assistants do not process data themselves. They write and fix the code that runs the stack below. You don't need all these tools for one project, but they combine perfectly.



Same Job, Different Author

Machine learning is simply the next step in logic. Instead of writing the classification rule by hand, the computer learns the rule from examples you have already labeled.

Human Authored



Expert
Experience



$Sv(120\text{ kHz}) - Sv(38\text{ kHz}) > \text{threshold}$



Classification

Machine Authored



Expert Labelled
Examples + Raw Data



Machine Learning Algorithm



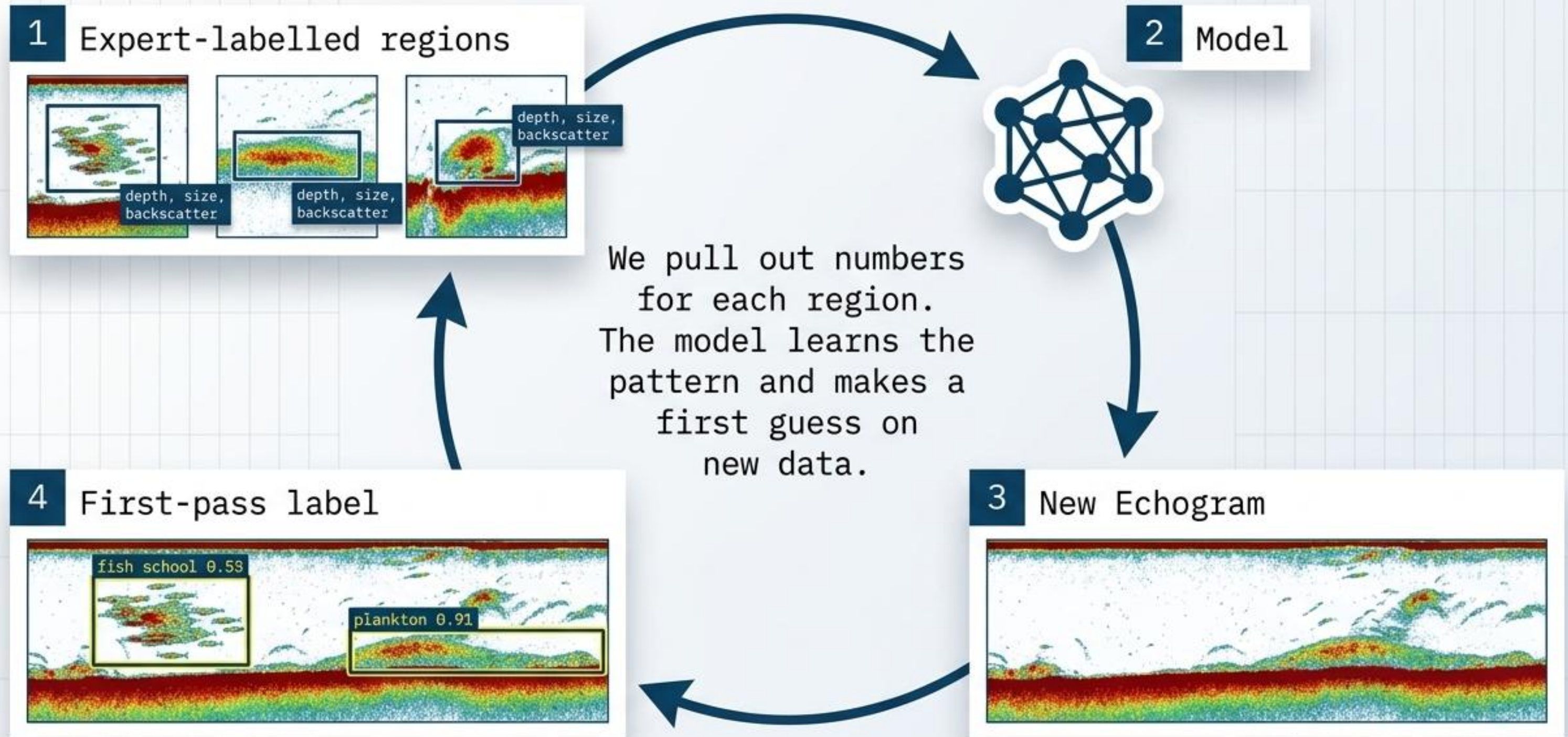
$Sv(120\text{ kHz}) - Sv(38\text{ kHz}) > \text{threshold}$



Classification



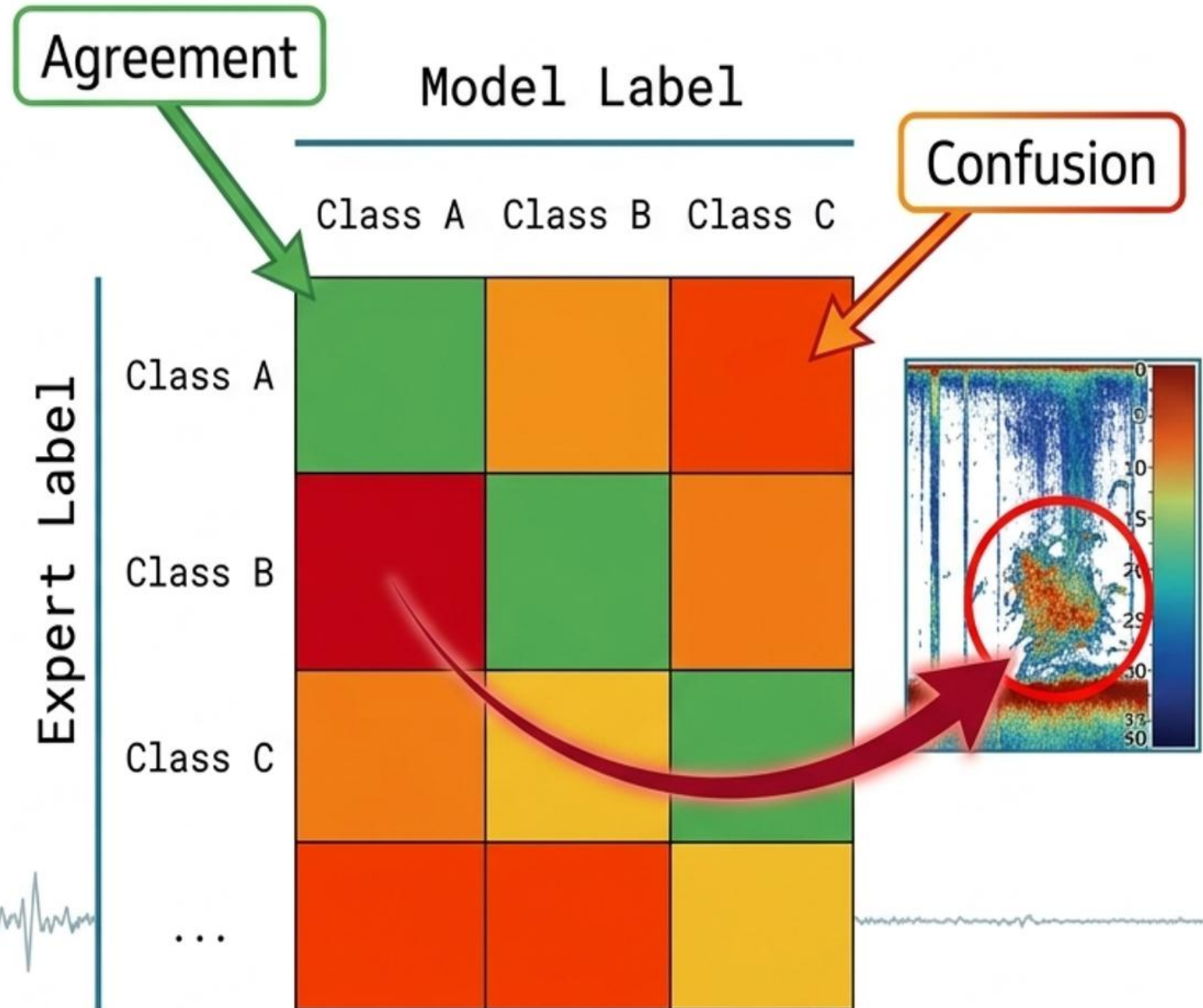
From echogram to prediction.



From Echogram to First-Pass Prediction

A model takes expert labels, learns the pattern, and makes a first guess on new data.

The confusion matrix isn't just an error report. The diagonal is agreement; everything off it tells the scientist exactly where to look next.



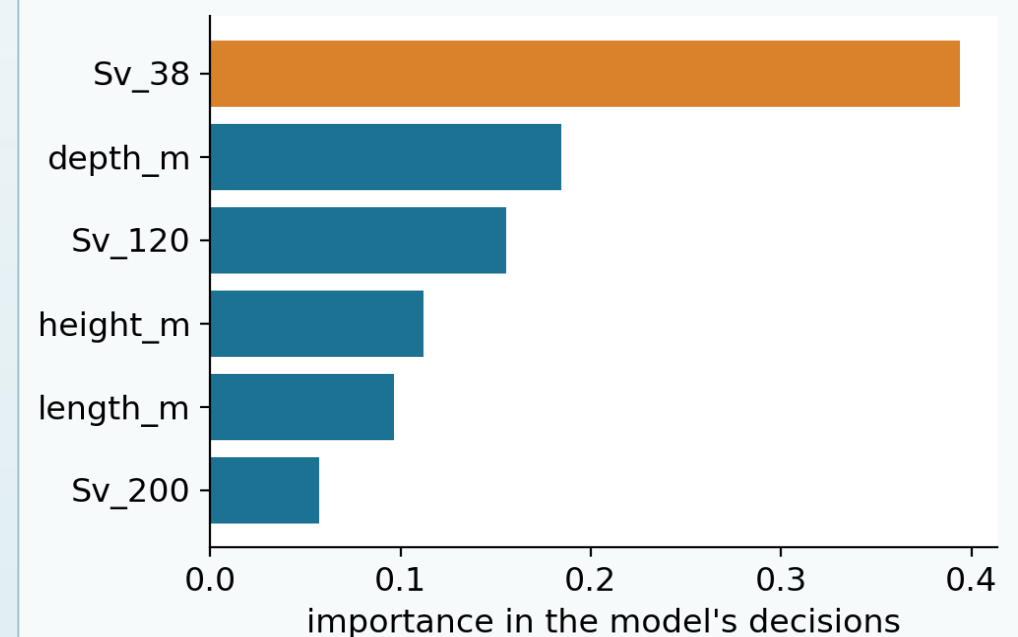


Live example 2 — from labelled schools to a first pass

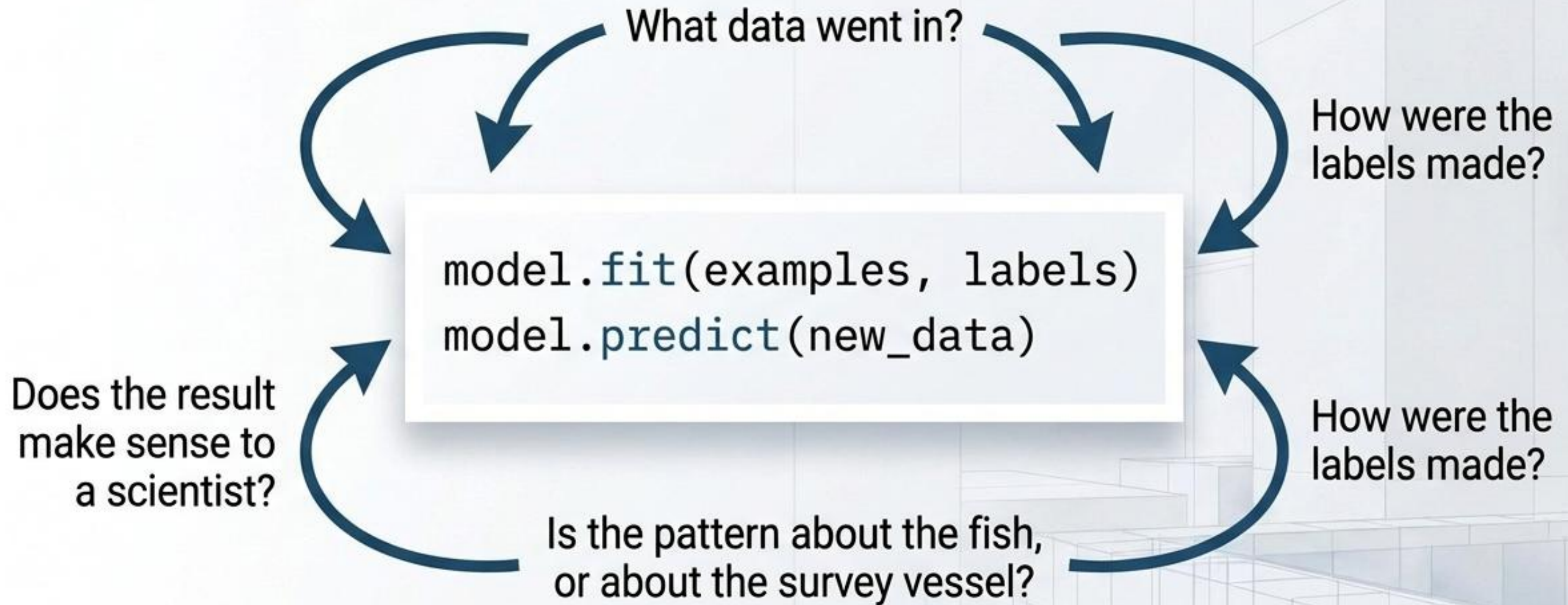
What I asked the assistant, *“Train a random-forest classifier to predict the species from the numeric columns, hold out 30 % of the rows for testing, and show the confusion matrix and the feature importances.”*

```
features = ["Sv_38", "Sv_120", "Sv_200", "depth_m", "height_m",  
            "length_m"]  
X, y = schools[features], schools["species"]    # one row per school  
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.3,  
                                           stratify=y)  
  
model = RandomForestClassifier(n_estimators=200)    # a  
demonstration model  
model.fit(X_tr, y_tr)                            # learn from labelled examples  
pred = model.predict(X_te)                        # first pass on unseen schools  
  
confusion_matrix(y_te, pred)                    # where it agrees, where it doesn't  
model.feature_importances_                      # which columns drive the decision
```

Which columns drive the decision?



What the model actually does.



The two lines of code are the easy part. Everything that makes the result trustworthy relies entirely on scientific judgement. First pass by the model. Uncertain cases back to the expert.

The Division of Labor

AI gives you a shorter list of things worth investigating. What those things mean is still your job.
AI can help us investigate our science; it shouldn't pretend to explain it.

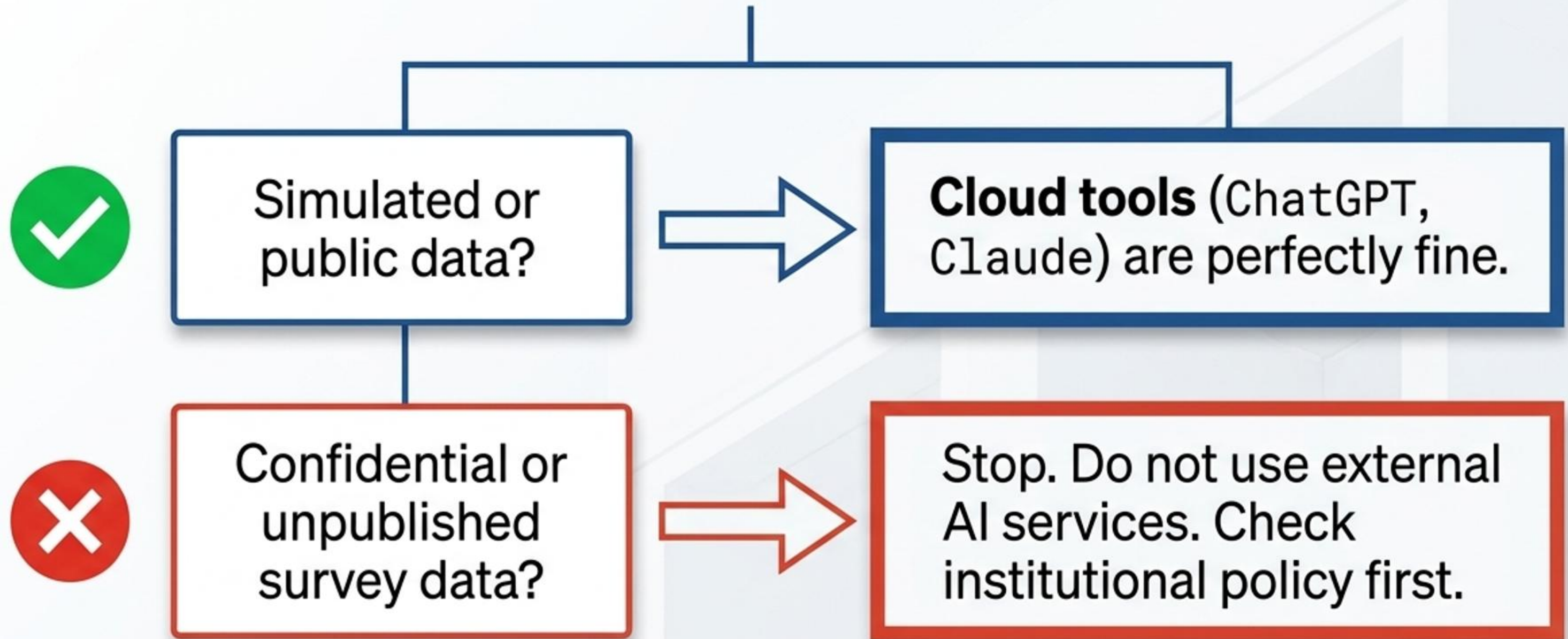
The AI Research Assistant

-  Processes large data
-  Finds similar regions at scale
-  Flags unusual observations
-  Generates figures
-  Automates pipelines
-  Makes first-pass predictions

The Independent Scientist (You)

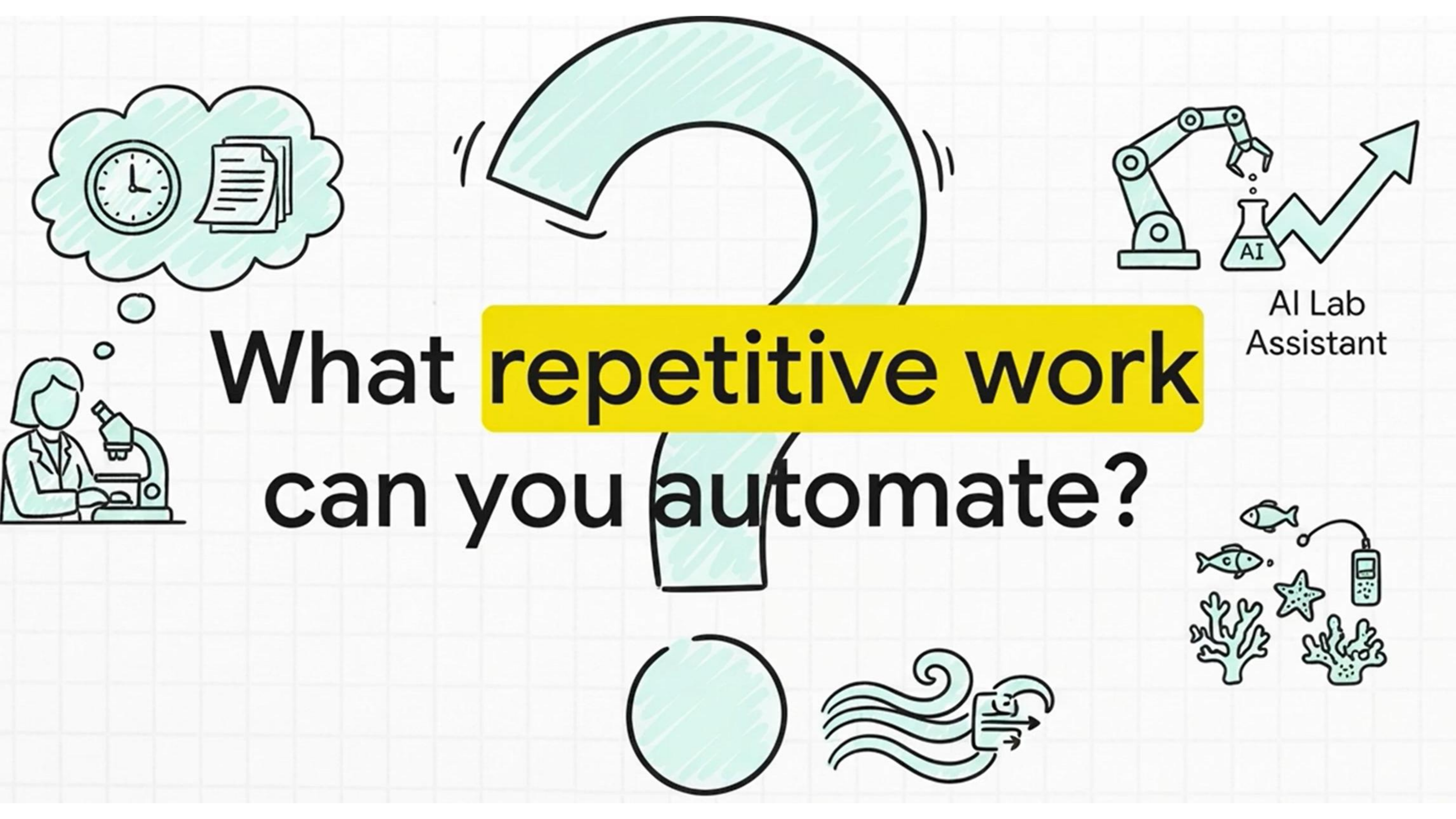
-  Defines the research question
-  Provides accurate labels
-  Determines if patterns are survey artifacts
-  Handles uncertain cases
-  Interprets scientific meaning

The Practical Rule for Data Privacy.



There are two distinct questions: **Can the technology do this?** and **Is it appropriate to do it this way with this data?**

The second question is yours to answer.



What **repetitive work**
can you automate?

AI Lab
Assistant

AI takes the repetition. We keep the judgement.

The goal isn't to replace your scientific workflow. It is to make parts of it faster, more reproducible, and easier to scale. Bring in machine learning only where it genuinely adds something.

Final Challenge

Which part of your current workflow is repetitive enough that you'd rather have a computer do it consistently than do it by hand every time?

(That is the right place to start)

Thank you!

Any Questions?



For my publications, projects and updates, scan the QR-code or visit
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(Hafró)